Optimasi Portofolio Menggunakan Variational Quantum Eigensolver (VQE)

Replikasi Studi Kasus pada Saham Teknologi

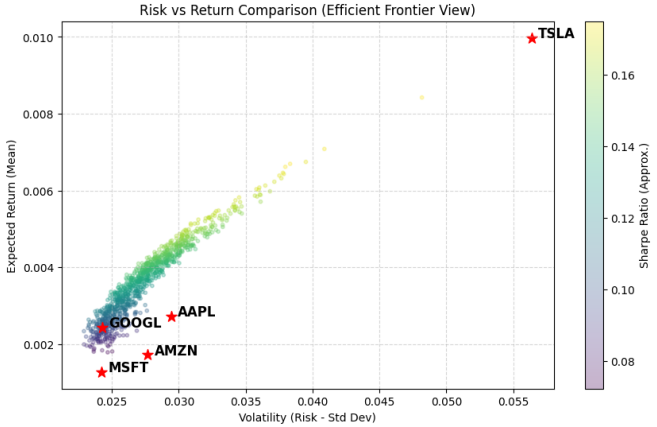
January 29, 2026

Outline

1. Pendahuluan
2. Formulasi Matematis
3. Mapping Kuantum
4. Studi Kasus
5. Implementasi VQE
6. Hasil

Pendahuluan & Latar Belakang

- **Masalah Optimasi Portofolio:**
  - Menyeimbangkan antara keuntungan (*return*) dan risiko (*risk*).
- **Pendekatan Klasik:**
  - Modern Portfolio Theory (Markowitz).
- **Pendekatan Kuantum:**
  - Menggunakan algoritma hibrid VQE (*Variational Quantum Eigensolver*).
  - Bertujuan menyelesaikan masalah kombinatorial yang kompleks.



Formulasi Matematis: Markowitz ke QUBO

1. **Fungsi Objektif Awal (Markowitz)**

$$\text{Minimize} \quad w \underbrace{\sum_{i,j} \sigma_{ij} x_i x_j}_{\text{Risk (Variance)}} - (1-w) \underbrace{\sum_{i} \mu_i x_i}_{\text{Return}}$$

- $x_i \in \{0, 1\}$: Variabel biner (pilih aset atau tidak).
- w: Parameter penghindaran risiko (*risk aversion*).

2. **Constraint (Kendala)**

$$\sum_{i=1}^{N} x_i = B \quad \text{(Memilih tepat } B \text{ aset)}$$

Formulasi Matematis: Metode Penalti

3. Metode Penalti (Unconstrained Optimization) Agar dapat diselesaikan dengan VQE, kendala diubah menjadi *penalty term*.

$$\text{Penalty} = P\left(\sum_{i=1}^{N} x_i - B\right)^2$$

4. Total Cost Function (Hamiltonian)

$$C(x) = wx^T\Sigma x - (1-w)\mu^T x + P\left(\sum x_i - B\right)^2$$

Mapping ke Hamiltonian Kuantum

- **Harus ada cara agar komputer kuantum memahami variabel biner**
- Mapping variabel biner $x_i \in \{0, 1\}$ ke operator spin/qubit Z_i.
- **Transformasi:**

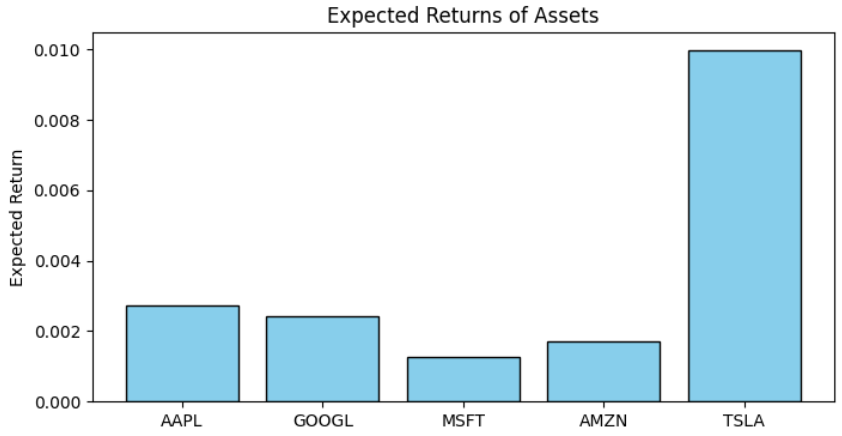
$$x_i \to \frac{1 - Z_i}{2}$$

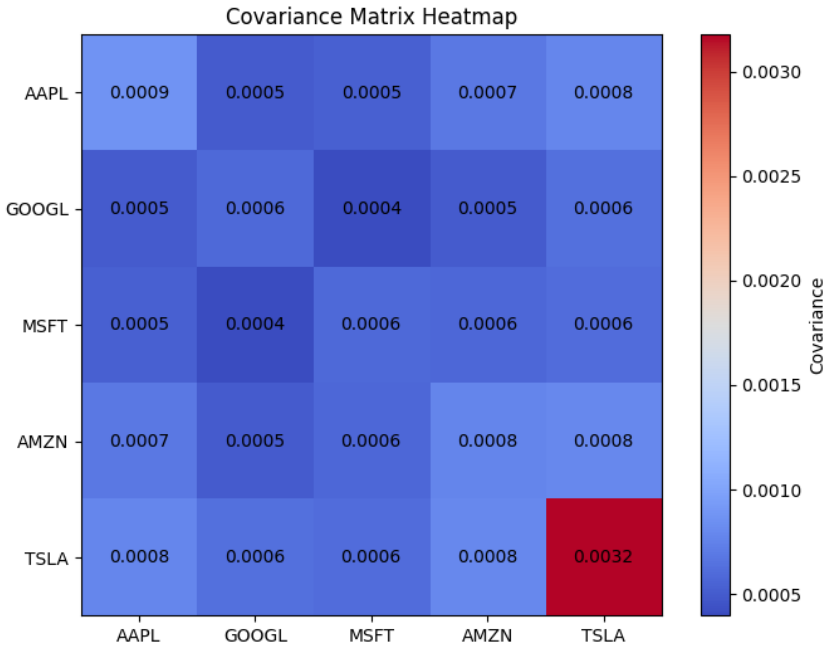
- Hasil akhirnya adalah **Ising Hamiltonian** yang akan diminimalkan energinya oleh VQE.

$$H = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j + \text{constant}$$

Persiapan Data & Studi Kasus

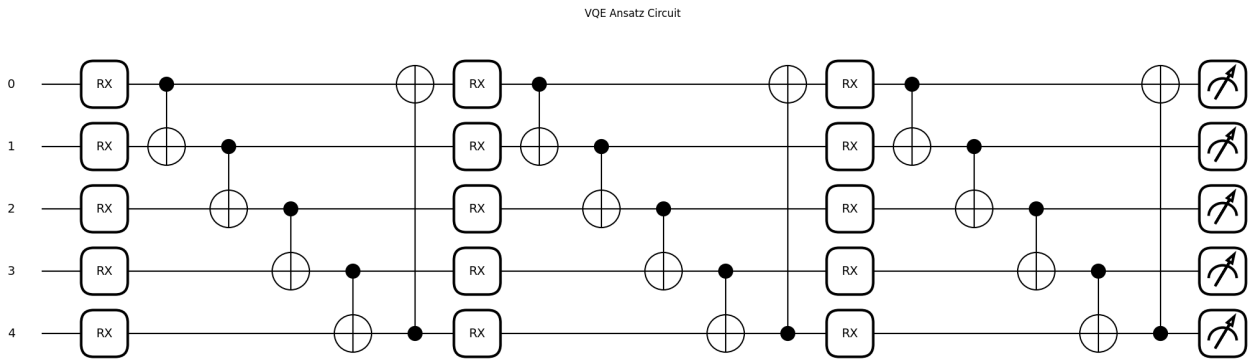
- **Aset:** 5 Saham Teknologi (AAPL, GOOGL, MSFT, AMZN, TSLA).
- **Sumber Data:** Yahoo Finance (2020-2021).
- **Parameter:**
  - Risk Aversion (w) = 0.5
  - Budget (B) = 2 aset

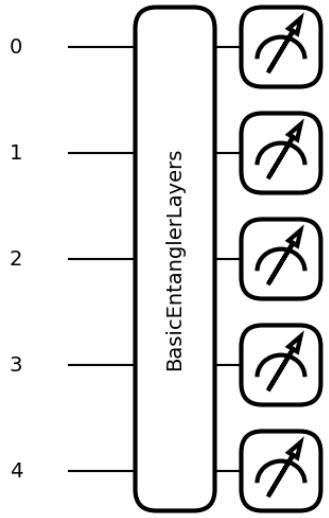




Implementasi VQE dengan PennyLane

- **Framework:** PennyLane.
- **Ansatz:**
  - Menggunakan *Hardware-efficient ansatz* (`BasicEntanglerLayers`).
  - Kedalaman: 3 layer.
- **Optimizer:**
  - Gradient Descent Klasik untuk mengupdate parameter sirkuit.
- **Proses:** Loop hibrid (Quantum Circuit $\leftrightarrow$ Classical Optimizer).

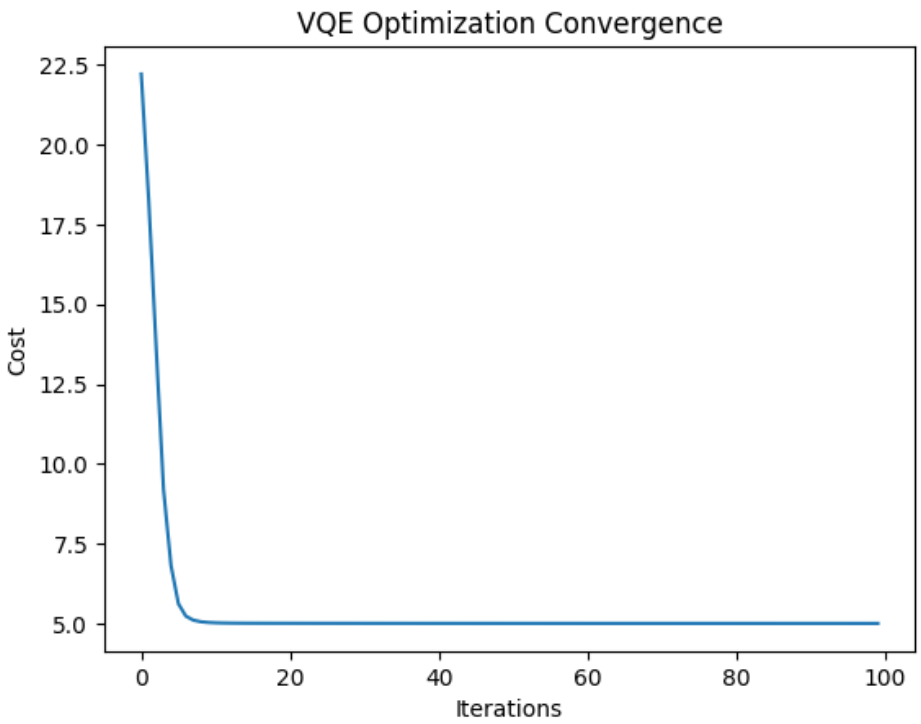




Hasil Optimasi

Grafik Konvergensi

- Penurunan nilai Cost function seiring iterasi (0 s.d 100).



Hasil Akhir

- Probabilitas State Akhir tertinggi = Solusi Optimal.
- **Optimal State:** `01001`
- **Aset Terpilih:**
  - GOOGL
  - TSLA